Problem Chosen	2024 MCM/ICM Summary Sheet	Team Control Number
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Catch the Flow of a Match with Improved LSTM

Summary

Firstly, based on the existing dataset, we used a Random Forest model to filter out the most significant attributes worthy of further investigation. The weights of these attributes were determined to facilitate subsequent analysis. Building upon the random forest model, we introduced a **quantitative model for psychological stress**, which utilized different indicators selected by the random forest model. **Leveraging relevant physiological and medical mechanisms**, this model effectively quantified the psychological stress experienced by tennis players under different conditions. Our model evaluation indicated that the serving side, current score, set, and match count had a significant impact on the psychological stress of players, with the serving side showing the strongest correlation, aligning closely with the mentioned concept of the serving side winning the game or achieving match points. This observation intuitively demonstrated the validity of our model.

Next, to more comprehensively quantify the impact of different scenarios on tennis players, we further supplemented the model with a physiological function quantification model as a complement to the psychological stress model. By considering the energy expenditure during movement, the impact of hitting styles on energy consumption, and the influence of receiving distance on physical stamina—these three core elements, combined with principles from physics, provided a thorough assessment of the current physical condition of tennis players. Integrating the psychological stress quantification model based on the **random forest**, we effectively addressed the challenging issue of describing "momentum" and transformed it into a score that is easy to analyze and compare.

Following that, based on the data provided by the first two quantification models, we proposed a **dual-layer Long Short-Term Memory (LSTM) model**. This model not only fully utilized the scores quantified by the first two models but also considered the significant differences in indicators affecting psychological stress and physiological function during different stages of the match. By alternately employing short-term memory and long-term memory models, it successfully predicted the upcoming player's "momentum." The predicted fluctuations in "momentum" not only countered the notion that fluctuations in a match and a player's success are random but also provided a theoretical basis for tennis coaches on how to optimize their serving strategies based on specific circumstances.

Subsequently, we conducted **sensitivity tests** on the existing models, and the results showed that the model exhibited a high degree of fit, particularly demonstrating significant sensitivity when handling key parameters such as server, set, score, and game. The test results not only showcased the model's adaptability to these parameters but also affirmed its outstanding ability to filter and respond to factors influencing match outcomes, further highlighting the model's validity.

Finally, we attempted to train the model to be a universal model applicable to various sports, but the results were not very promising. We propose corresponding directions for improvement.

Keywords: Psychological Stress Quantification Model based on Random Forest; Physiological Function Quantification Model; Dual-Layer Long Short-Term Memory (LSTM) Model

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1 Introduction

1.1 Problem Background

Momentum is frequently employed in daily life, as exemplified by the physics definition: the force that keeps an object in motion. Similarly, when watching tennis matches and adhering to scoring rules as illustrated in Figure 1, it becomes apparent that players who dominate the game often evoke the expression ”having momentum.”

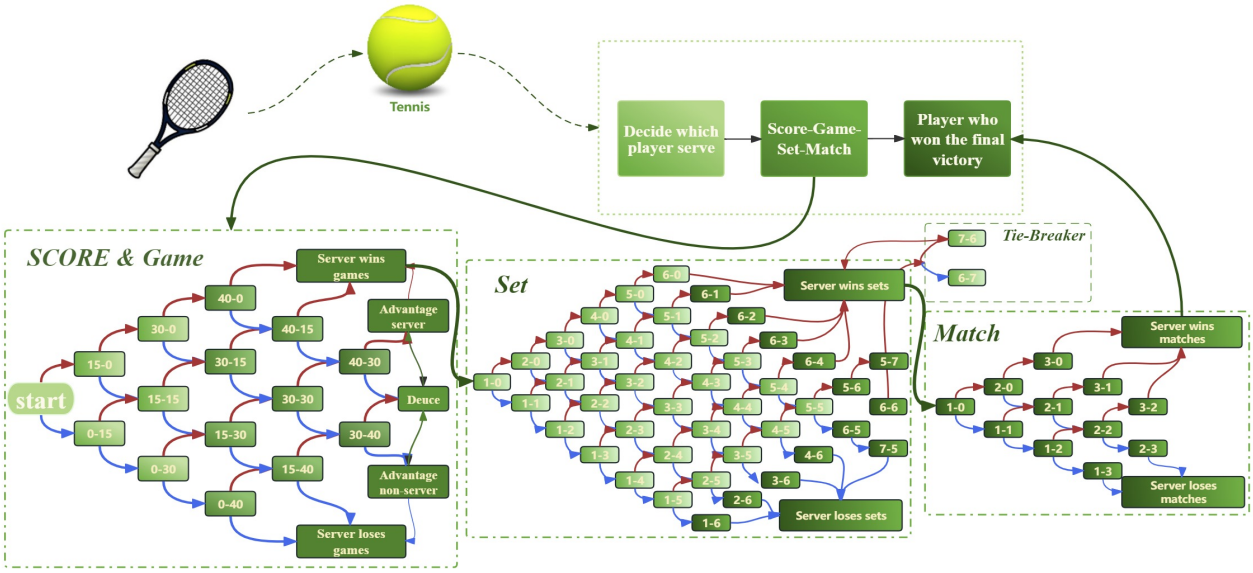


Figure 1: Tennis Scoring Rules [2, 3]

In tennis, the concept of momentum takes on a completely different meaning: the psychological and strategic advantage gained by a player or team after a series of successful matches or points. Momentum plays a crucial role in shaping the dynamics of tennis matches. It not only influences the mindset of individual players through positive cognitive shifts, including changes in self-efficacy, motivation, and attention [1], but also impacts the course of the game.

According to the general feelings of people on a daily basis, one key aspect of momentum in tennis is its impact on a player's confidence. Psychological Momentum (PM) has been proven to exist, which refers to the ”power” or ”energy” a player has during a game. [4].Researchers pay particular attention to its performance and role at key turning points in the game, such as at match point, when a lagging player unexpectedly reverses the situation [5].

Therefore, from a strategic perspective, coaches consistently formulate strategies by anticipating the momentum shifts in tennis matches. This is a crucial rule influencing the trajectory of the game. Players with momentum often dictate the pace of the match, applying pressure to their opponents, forcing them into defensive positions, and potentially triggering a domino effect.

1.2 Clarification and Restatement

Our mathematical model is tasked with intricate computations and detailed analyses, ensuring the completion of multifaceted objectives. Through this, it aims to deliver comprehensive solutions and profound insights. Following are the tasks our mathematical model needs to complete:

Task1 Develop a model to capture the flow of scoring in tennis matches and apply it to one or multiple games. The model should identify players who perform better at specific moments in the match and showcase their performance advantages. Provide a visualization based on the model to illustrate the progression of the game.

Task2 Use the model to assess a tennis coach's perspective who doubts the role of "momentum" in matches, believing that changes and winning streaks in games are random.

Task3 Develop a model that predicts momentum shifts in a match based on data from at least one game. Investigate which factors are most relevant, and propose strategic recommendations for players to counter different opponents based on past variations in momentum.

Task4 Test the developed model on other matches, evaluate its effectiveness in predicting changes in games, explore factors that may need inclusion in case of poor model performance, and assess the model's applicability to other matches, tournaments, court surfaces, and different sports.

1.3 Our Work

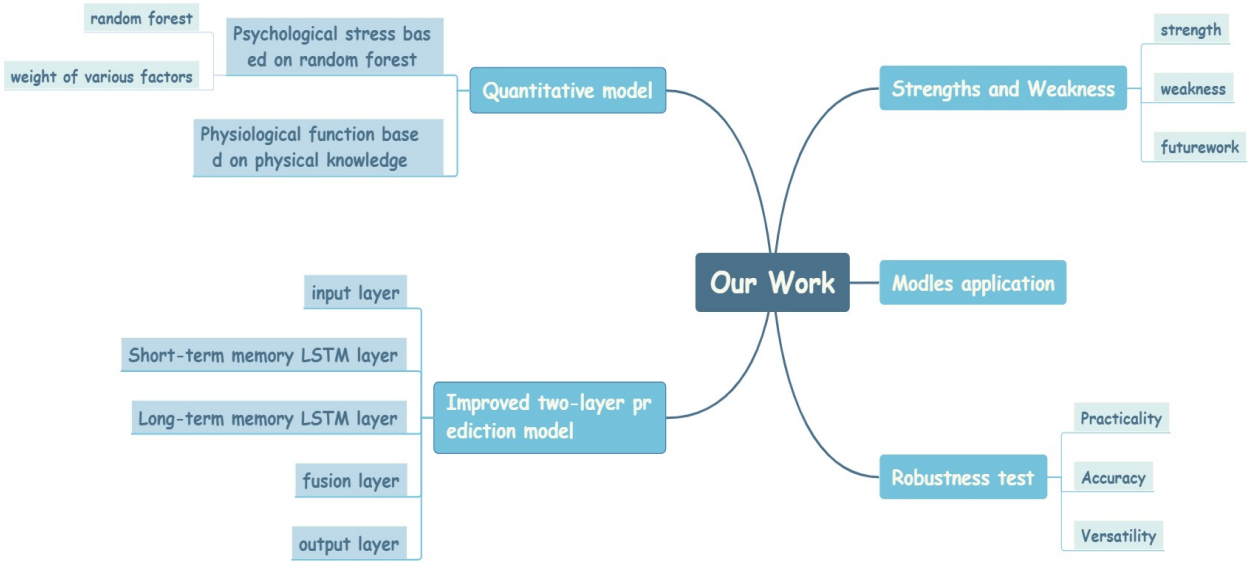


Figure 2: Our Work

2 Data Preprocessing and Preparation of the Models

2.1 Preparation of the Models

In this study, we initially conducted meticulous preprocessing of tennis match data, aiming to establish a solid foundation for subsequent dynamic model analysis. Data preprocessing comprised two primary steps: outlier handling and missing value treatment.

By applying box plots and the Interquartile Range (IQR) method, we identified and addressed outliers in the dataset to eliminate their potential negative impact on the model's performance. Subsequently, various techniques were employed to address missing values, including the removal of variables or samples with substantial missing data, as well as imputation using mean, median, or model-based approaches for data with fewer missing values.

2.2 Data Processing

These preprocessing steps as fig.3 shows ensured the quality and consistency of the data, providing essential support for constructing an efficient and accurate predictive model.

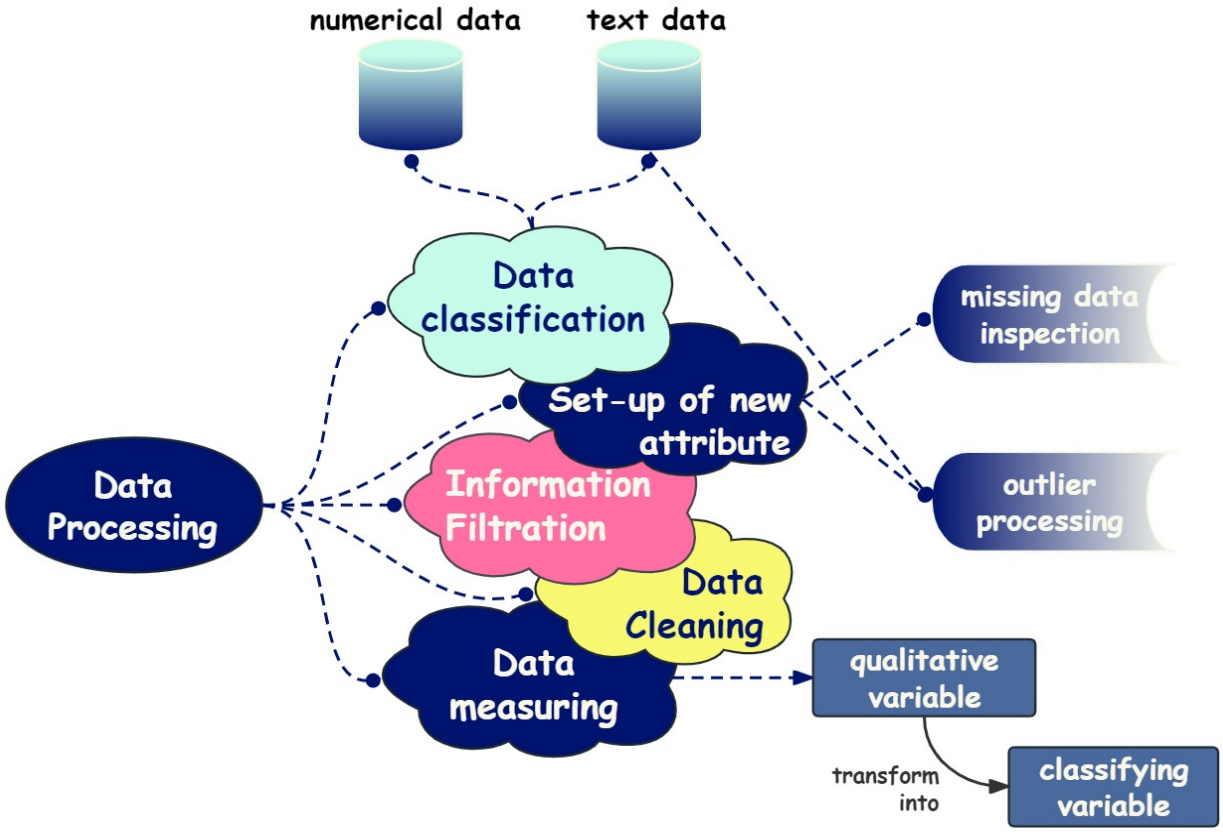


Figure 3: Data Preprocessing

3 Model Construction

3.1 Psychological Stress Quantification Model based on Random Forest

3.1.1 Weight Analysis

The reason why scores and match points affect a player's "momentum" is fundamentally due to their impact on the player's psychological state. Therefore, we need to analyze in detail how these indicators affect the "momentum," to what extent they influence the "momentum" and quantify the corresponding data for the subsequent handling of the fluctuation prediction model.

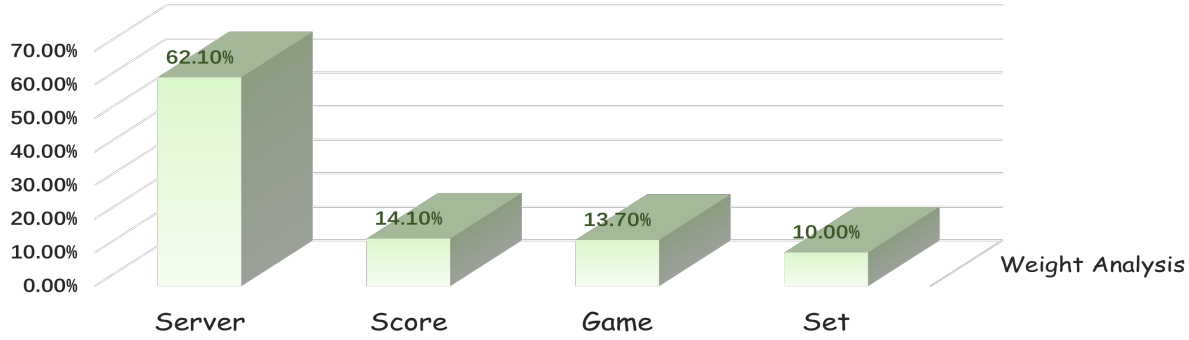


Figure 4: The weight of various factors on psychological stress

Thus the group assess the players' psychological pressure based on the current overall situation on the court. To establish a quantifiable standard for psychological pressure, we aimed to calculate the weights of different factors.

Simplifying the problem, we ultimately selected four factors: the serving side and the current scores in the match (score), game, and set. We transformed the scores into nominal labels, simulating the scores as meaningful symbols that significantly impact the players. Subsequently, we employed a random forest model to calculate the relative importance of these four factors in determining the outcome of the upcoming point.

We will use these four ratios as weights in the subsequent steps to quantify psychological distress. They are respectively w_1, w_2, w_3, w_4.

Table 1: Model Evaluation

	MSE	RMSE	MAE	MAPE
Training set	0.206	0.454	0.423	28.761
Test set	0.218	0.467	0.433	29.35

This is the result of model evaluation. We can see that the values of MSE, RMSE, etc. are relatively small, indicating that the accuracy of the model is high.

3.1.2 Model Establishment

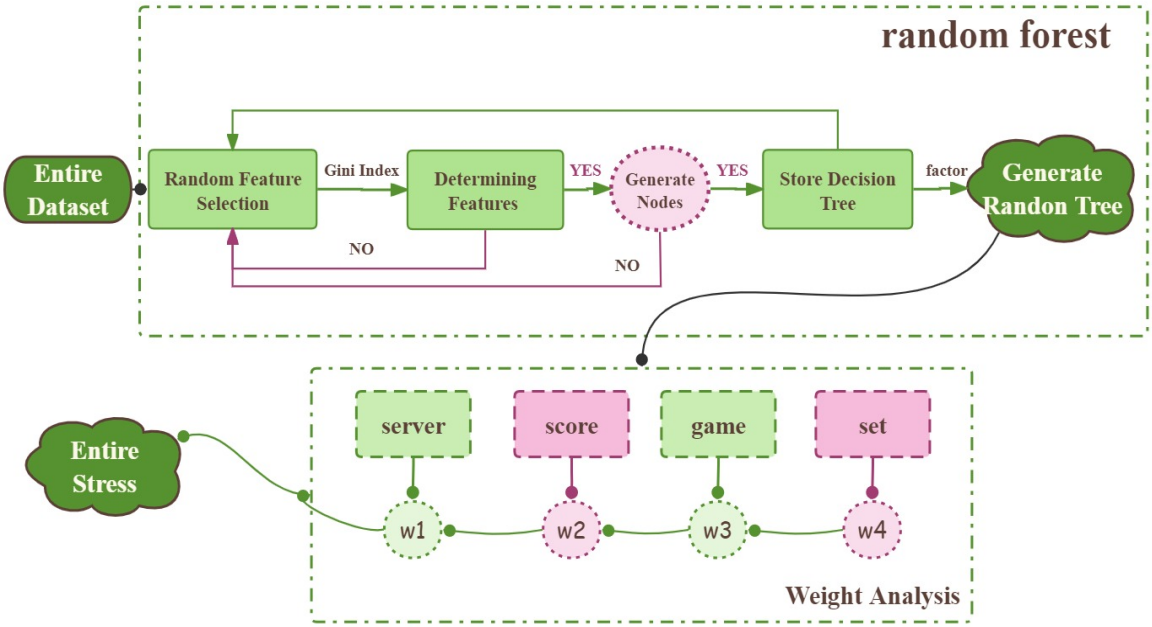


Figure 5: Random Forest Flow Chart

Based on above, we constructed a psychological stress quantification model that combines four key factors output by the random forest model: server, current score, game, and set. The selection of these factors is based on their importance in the game and their significant impact on the athletes' mental state. The model is quantified by the following formula:

$$P_{stress} = w_1 \cdot P_{server} + w_2 \cdot P_{score} + w_3 \cdot P_{game} + w_4 \cdot P_{set} \tag{1}$$

Among them, P_{server}, P_{score}, P_{game} and P_{set} respectively represent the server, current score, game number and set. The contribution of numbers to athletes' psychological pressure is 62.1%, 14.1%, 13.7% and 10% respectively. This reflects the different effects of these factors on players' mental states during tennis matches. The identity of the server is the most important factor because it directly affects the scoring probability of the game. The other three factors reveal pressure changes during the game from different dimensions of the game. This model is designed to provide coaches and athletes with

precise psychological stress management strategies to help them better understand and respond to the psychological challenges of competition.

3.2 Physiological Function Quantification Model

3.2.1 Physiological Factor Analysis

We noticed that not only the psychological state but also the physical state influences the player's "momentum." Therefore, while keeping the indicators affecting the psychological state unchanged, we simulated which indicators still have an impact on predicting the results of the next game. We selected these indicators to quantify our analysis of the physical state.

Thus, our quantitative analysis of physiological function focuses primarily on three aspects: the energy expenditure related to running distance, the energy consumption associated with hitting speed, and the slight impact of receiving distance on stamina. By integrating these factors, we developed a model to assess the physical performance of tennis players. The energy expenditure of running distance is estimated based on the player's movement distance and speed during the match. The energy consumption of hitting speed is calculated based on the speed and frequency of shots. The impact of receiving distance considers the distance a player needs to move to receive a shot and its effect on stamina.



Figure 6: Physical Endurance

3.2.2 Model Establishment Based on METs

We used a comprehensive mathematical model to quantify the physiology of tennis players. First, the physical energy consumption of the running process is evaluated through metabolic equivalents (METs). Metabolism is calculated through the following formula

$$E_{run} = METs \times Weight \times Time \tag{2}$$

where E_{run} represents Energy consumption during running, $METs$ is the metabolic equivalent value determined based on activity intensity, $Weight$ is the athlete's weight, and $Time$ is the exercise time. The energy cost of batting speed is estimated by the following formula:

$$E_{stroke} = \frac{1}{2}mv^2 \tag{3}$$

where m represents the force used in hitting the ball (taking into account muscle strength input), v is the speed of the ball.

Finally, the impact of catch distance on physical performance was evaluated using a simplified physical activity model that takes into account the additional energy required by the athlete to move to the catch position. Combining these three aspects, we calculate the final physical fitness score S through a weighted average, the formula is

$$S = w_1 E_{run} + w_2 E_{stroke} + w_3 E_{receive} \tag{4}$$

where w_1, w_2, w_3 is the weight based on the impact of each factor on the game.

3.3 Dual-Layer Long Short-Term Memory (LSTM) Model

We found that when the score difference is large or small, the players' mentality differs significantly, which is directly reflected in the number of mistakes and receptions. Therefore, we cannot predict the states of all time periods using a single model, as this leads to significant errors. Consequently, we developed Dual-Layer Long Short-Term Memory (LSTM) Model.

When designing the model to predict the performance of tennis players in matches, we employed a dual LSTM layer strategy to handle dependencies at different time scales. This model structure aims to capture both short-term dynamics (e.g., changes during point stages) and long-term trends (e.g., trends throughout the entire set stage) in the match.

3.3.1 Model refinement

- **Input layer:**

  For each time point t, the model receives a feature vector X_t, containing current match-related information, covering physiological and psychological indicators of the athletes. The purpose of this layer is to transform real-time match data into a format that the model can process.

- **Short-term memory LSTM layer:**

  - This layer is designed to capture real-time changes in the match, such as the outcomes of individual points or games.

  - The update mechanism employs a sophisticated gating mechanism, calculated through

$$H_{st} = \sigma(W_{st} \cdot [H_{st-1}, X_t] + b_{st}) \tag{5}$$

  where σ represents the sigmoid activation function responsible for regulating the degree of opening for the gate units. W_{st} and b_{st} respectively denote the weight matrix and bias term, which are parameters optimized through the learning process.

- **long Term Memory LSTM Layer:**

  - Focused on understanding the progress of the game from a broader perspective, such as the overall trends throughout the entire match.

  - This layer, through modeling long-term dependencies, updates its state with

$$H_{lt} = \sigma(W_{lt} \cdot [H_{lt-1}, X_t] + b_{lt}) \tag{6}$$

  employing the sigmoid function in a similar manner to control the flow of information.

- **Fusion Layer:**

  By combining the outputs of the short-term and long-term memory layers, this layer generates a comprehensive representation of the match state:

$$C_t = \alpha \cdot H_{st} + (1 - \alpha) \cdot H_{lt} \tag{7}$$

  where α is a pre-defined parameter used to balance the importance of short-term and long-term outputs.

- **Output Layer:**

  - Based on the fused representationC_t, , the model generates performance scores for the athlete at time t through

$$Y_t = softmax(W_y \cdot C_t + b_y) \tag{8}$$

  Here, the softmax function ensures that the final output forms a probability distribution, allowing us to interpret the uncertainty in the model's predictions.

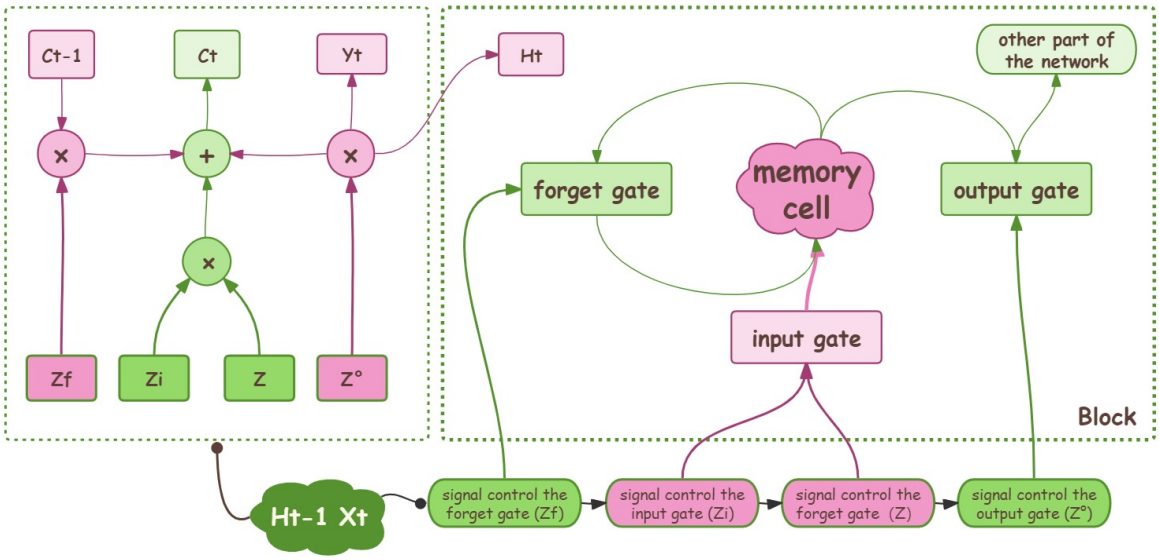


Figure 7: LSTM model

3.3.2 Model application

- In the point stage of the match, the short-term memory LSTM layer plays a more significant role because rapid changes in the game during this phase have a substantial impact on immediate performance.
- As the match progresses into the game and set stages, the long-term memory LSTM layer becomes more crucial, leveraging the overall process and trends of the game to predict the player's performance.
- Through this structure, the model can flexibly adapt to different stages of the match, providing coaches and players with accurate performance predictions and a basis for strategic adjustments.

Through the aforementioned steps, our model combines the strengths of both short-term and long-term memory layers, providing a comprehensive analysis and prediction of a tennis player's performance during a match. The proposed dual-layer LSTM model exhibits significant advantages over the original single-layer model, particularly when dealing with tennis match data with varying time-scale dependencies.

Firstly, the short-term memory layer captures immediate changes in the match, such as winning a single point, providing the model with sensitivity to the current match situation. For instance, in the point and game stages, short-term score impacts are more significant, while long-term effects are minimal. Additionally, at the early stages of the game, players often focus on later matches, making short-term memory appropriate.

Secondly, the long-term memory layer focuses on capturing the overall trends in the match and the long-term changes in player performance, such as the progress of the entire set or match, providing the

model with a deep understanding of context. For example, after the conclusion of the first set, when it becomes challenging to catch up in score, players may experience increased pressure. At this point, it is essential not to overlook the performance in the earlier stages, making long-term memory appropriate.

This structure allows the model to simultaneously consider short-term and long-term factors when predicting a player's performance, enhancing prediction accuracy and robustness. Additionally, through fine-tuning the interaction between the two layers, our model adapts more flexibly to dynamic changes in the match, providing coaches and players with more valuable insights.

Thus We used a two-layer LSTM model to predict fluctuations in the quantified indicators, as shown in the graph below. We compared the predicted trends with actual data from individual matches, and the results indicated a close match between the two.

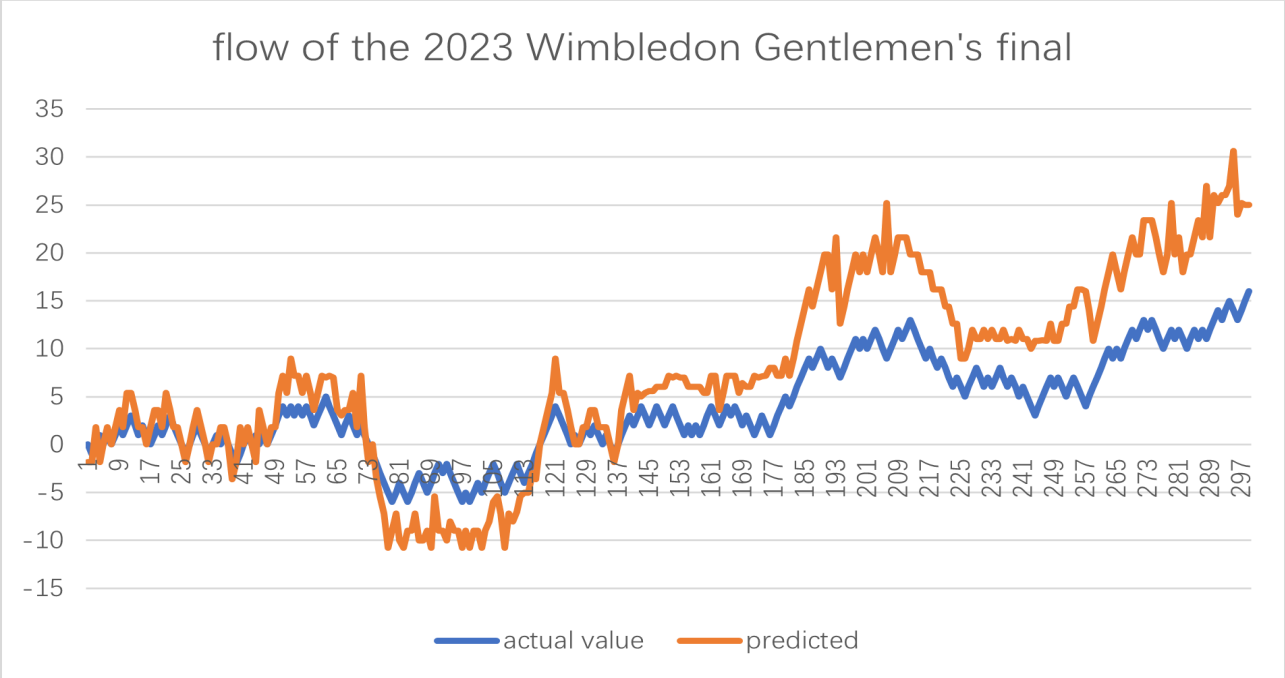


Figure 8: Flow of the Match

4 Model-based Strategy Analysis

4.1 Feature Importance Analysis

To delve deeper into the volatility within matches and its correlation with various factors, this study introduces a series of factors influencing match outcomes. The random forest classification method is applied to assess the importance of these factors in determining the outcomes of individual game sets.

Table 2: Feature Importance

Feature	Feature Importance
total shots in each point	43.80%
server	28.60%
p1's running distance	10.70%
p2's running distance	10.00%
score	2.80%
game	2.60%
set	1.50%

In contrast to previous analyses focusing solely on the current situation, this study meticulously examines the specific performance of players in the preceding set and its potential impact on the upcoming match. Surprisingly, we found a significant correlation between the performance in the previous set, such as the total number of shots and running distance, and the results of the subsequent set. This discovery suggests that recent events in the match have a more substantial impact on players than events from a more distant past, reflecting players' heightened attention to recent score changes rather than the overall score.

Through in-depth analysis, we also observed that regardless of score fluctuations, the role of the serving side remains consistently crucial in determining the outcomes of close-score situations in individual game sets. This finding underscores the critical role of serving in match strategy and emphasizes the need for players to optimize their serving strategies with specificity during the match.

4.2 Strategy Analysis

Based on these analytical results, we conclude that match volatility is closely associated with a player's performance in the previous set (e.g., total shots and running distance) and match strategy (such as the choice of the serving side). In light of this finding, we recommend:

- **Initiative:**

  Coaches and players should place greater emphasis on technical initiative and creativity, leveraging their unique playing styles to respond to match uncertainties and volatility.

- **Strategic Adjustments:**

  Players should flexibly adjust match strategies based on the immediate set situation and performance in the previous set, especially during serving and receiving, to exploit opponents' weaknesses and maximize their own advantages.

- **Data-Driven Training:**

  Coaching teams should utilize match data analysis to identify key factors influencing match outcomes and develop personalized training plans to enhance players' ability to navigate match volatility.

Simulated analyses were conducted on a series of matches using the trained model to predict win-loss results under specific conditions. The results were visually presented in the form of a heatmap. Confusion matrix analysis indicates a high level of accuracy in the simulated results, reflecting the effectiveness of the model in predicting match outcomes.

5 Sensitivity Analysis

5.1 Sensitivity Testing

We simulated a series of games using a trained model designed to predict the outcome of a given score in a specific situation. We visually displayed the simulation results in the form of heat maps. Analysis of the confusion matrix shows that the accuracy of the simulation results is high, reflecting the effectiveness of the model in predicting the outcome of the competition.

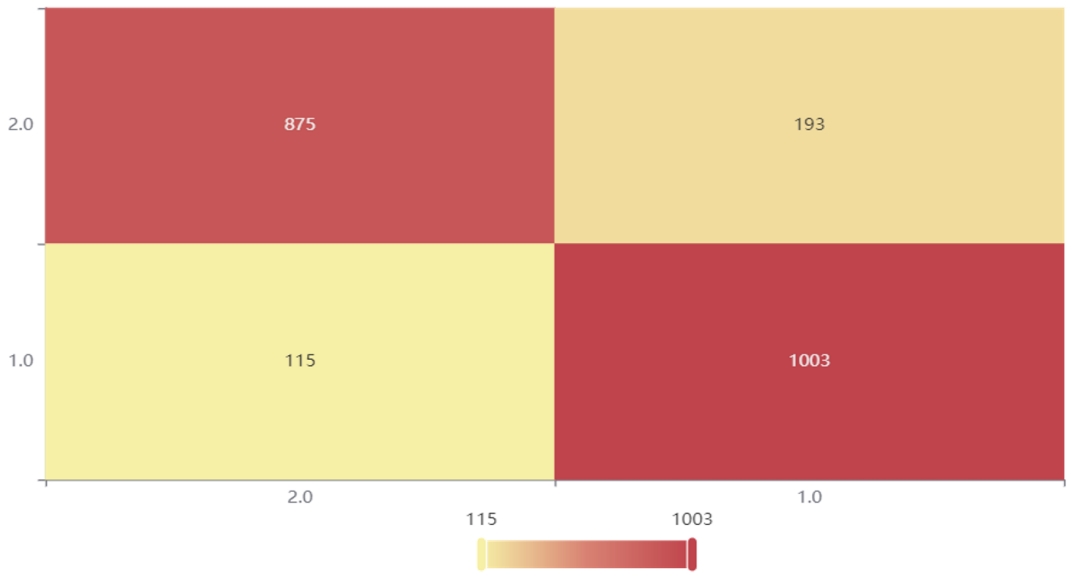


Figure 9: Heat map

We conducted detailed sensitivity testing on our model using the data from the "Wimbledon _ featured_ matches.csv" file. Through carefully designed experiments, we compared real win-loss data with model predictions and visualized this information in scatter plots. The results indicated a high level of fitting accuracy for the model, demonstrating its strong predictive capabilities for match outcomes.

Furthermore, we selected four key parameters server, set, score, and game and tested the model by inputting them in various combinations. The results of this process showed that the model exhibited high adaptability to different parameter combinations, with the highest sensitivity observed for the server parameter, followed by set, score, and game parameters. Testing with other parameters revealed their relatively minor impact on the model's output, further confirming the model's robustness and highlighting its accurate judgment in screening important influencing factors.

The table 3 below shows some specific details of the simulation process.

Table 3: Feature Importance

Predicted result Y	Point victor	Probability of prediction 1.0	Probability of prediction 2.0	server	set	score	game	P1's distance	P2's distance	Total shots
2.0	2 .0	0.167	0.833	2	4	4	25	0.811	0.784	1
2.0	2.0	0.280	0.720	2	4	18	25	14.406	14.676	3
1.0	1.0	0.627	0.373	1	4	1	9	22.374	25.757	7
1.0	1.0	0.863	0.137	1	4	12	9	1.468	1.859	1
1.0	2.0	0.6141	0.386	1	4	13	9	15.683	19.131	6
1.0	1.0	0.813	0.187	1	4	9	9	8.48	11.579	3
1.0	1.0	0.731	0.269	1	4	14	9	24.876	36.244	3
1.0	2.0	0.553	0.447	1	1	1	1	22.251	22.48	6
1.0	1.0	0.555	0.445	1	1	2	1	36.93	33.856	9
1.0	1.0	0.770	0.232	1	1	3	1	8.062	9.631	3
1.0	1.0	0.623	0.377	1	1	9	1	16.065	14.496	5
2.0	2.0	0.418	0.582	1	1	14	1	5.643	6.899	2
1.0	1.0	0.535	0.465	1	1	6	1	26.64	20.464	5
2.0	2.0	0.177	0.823	2	1	1	2	3.32	2.485	1
1.0	1.0	0.566	0.434	2	1	2	2	15.394	18.588	4

5.2 Sampling Simulation Test

We randomly selected data from several matches for simulation and chose two matches at random. Fifty consecutive data points were extracted from each of the two matches to create the scatter plot below. To visually represent the accuracy of our model, we set y to 0 and 1 to represent actual losses and wins, respectively. Values distributed between 0 and 1 represent the model's estimated probability of the player winning the match. In other words, the closer the points in the middle are to the points representing the actual outcomes, the higher the accuracy of our model.

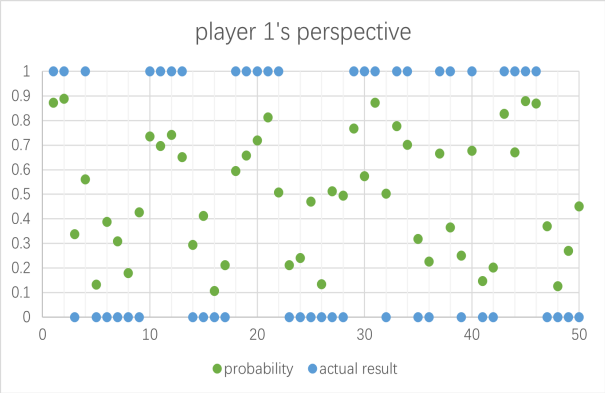


Figure 10: Player1

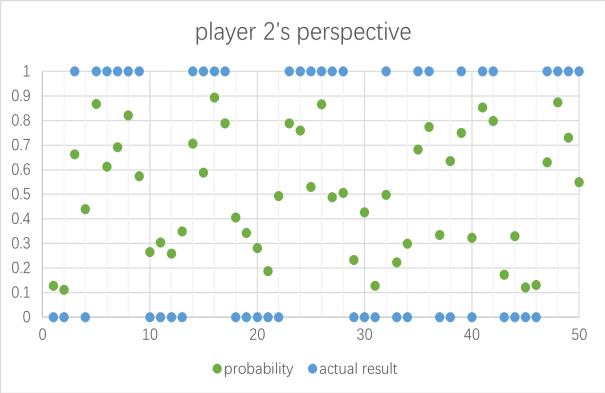


Figure 11: Player2

Figure 12: Match 1

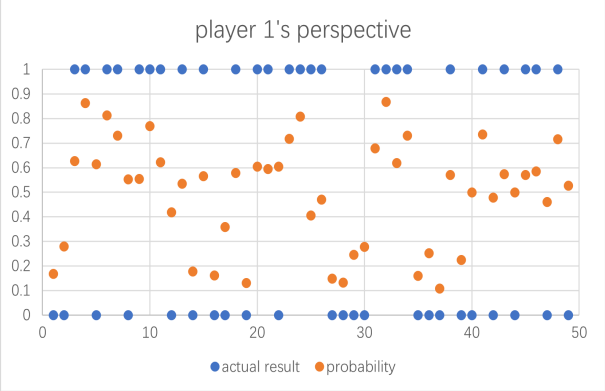


Figure 13: Player1

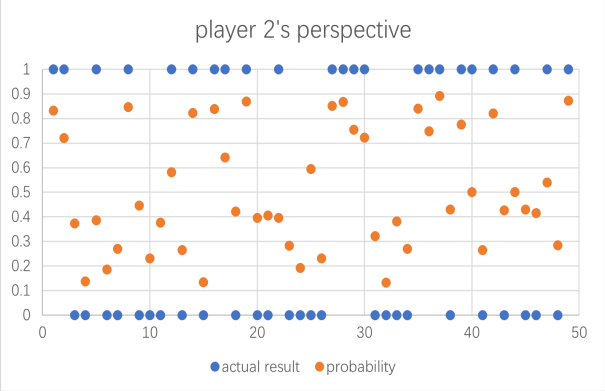


Figure 14: Player2

Figure 15: Match 2

6 Strengths and Weaknesses

6.1 Advantages Analysis

6.1.1 Strengths

- By utilizing the functionalities of dynamic models like Random Forest model and dual-layer LSTM, the model becomes empowered to intricately identify subtle shifts and recognize enduring patterns within the game. This heightened capability not only significantly enhances the understanding of performance fluctuations but also improves the accuracy and effectiveness of predicting these fluctuations in a nuanced and comprehensive manner.

- In the scope of this research, we meticulously developed an all-encompassing model employing random forest to meticulously analyze extensive data. This model seamlessly integrates quantification models for psychological pressure and physiological functions. The resulting framework provides a comprehensive performance assessment, adeptly forecasting the performance of tennis players with a holistic approach.

- Featuring a distinctive structure, the model excels in intricately analyzing players' psychological and physiological states during matches. Its dual-layer long short-term memory model showcases exceptional accuracy and sensitivity in predicting match momentum. Consequently, the model proves invaluable in devising tailored training plans and game strategies, offering substantial assistance in optimizing player performance.

6.1.2 Weaknesses

- Despite the model's excellent performance in handling key parameters and predicting momentum fluctuations, it has some limitations. It requires a substantial amount of historical data for model training, posing challenges in data collection and processing.

- The model's reliance on a vast dataset demands careful handling of outliers and missing values, limiting its application in data-limited settings. Addressing data quality challenges is crucial for broader usability.

- The intricate nature of the model might result in elevated computational expenses, heightened complexity in interpretation and application, and potential hurdles in conducting real-time analyses.

6.2 Future Work

In an ideal scenario, we would like to apply our model to a broader range of sports. However, due to the way we quantify "potential energy," our model incorporates many characteristics specific to tennis matches. These include the format differences compared to sports like table tennis or badminton, as well as variations in the characteristics of physical exertion during matches (e.g., the mileage covered in table tennis is much less than in tennis). When applying our model to other types of sports, it is necessary to reconsider the specific characteristics of each sport. This may make our model less generalizable.

These models provide insights into tennis performance fluctuations, enhancing decision-making in training and competition by integrating dynamic factors. Strengths include holistic performance assessment, but reliance on extensive data and complexity limit applicability. Future research should optimize structures and reduce data reliance, holding promise for revolutionizing tennis decision-making and advancing sports analytics.

7 Conclusion

In conclusion, the developed model offers coaches a valuable tool for predicting players' physical conditions based on real-time factors and digital metrics during a match. This innovative approach enables coaches to make informed decisions and strategically adapt game plans according to the dynamic situations on the field. By leveraging the insights provided by the model, users can proactively adjust player conditions, optimizing performance and minimizing the risk of fatigue or injuries. This not only enhances the overall effectiveness of coaching strategies but also contributes to the well-being and longevity of the players. The integration of data-driven predictions into coaching practices marks a significant advancement in sports management, fostering a more responsive and adaptable approach to player development and team dynamics. As technology continues to play a pivotal role in sports, the model stands as a testament to the potential for intelligent systems to revolutionize the way coaches analyze and optimize player performance in real-time scenarios.

Memo

To: Coaches and Athletes
From: Team #2400151
Date: February 5, 2024
Subject: The impact of "potential energy" on turning points and related suggestions

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Dear Coaches and Athletes,

In sports competitions, we can often see some exciting turning points, such as at match point, when the trailing side suddenly reverses the situation and wins the game. This is also the charm of sports competitions. Although athletes' performance on the field will be affected by very complex factors, researchers still hope to discover certain patterns or use scientific methods to study player performance. It is in this context that the word "momentum" is used in sports competitions, attracting the interest of many researchers. It describes a player's motivation or energy that is affected by a series of positive or negative events during the game. Our goal is to build a model to study the rules of the game when match points occur and predict the turning point of the game. At the same time, through our research, we are also very happy to provide you with some useful suggestions.

First, we use the algorithm that combines the random forest model and the weighted average to filter out the factors that have the greatest impact on the game results. After comprehensively considering factors such as the server, the current score, the number of games and the number of sets, we use different weights to Psychological stress is quantified into numerical values. On this basis, we further supplemented the physiological function quantification model, taking into account the three core elements of the physical consumption of running process, the different energy consumption of batting methods, and the impact of catching distance on physical strength, to evaluate Athletes' physical condition. Combining the two factors of psychological pressure and physical condition, we can evaluate the "momentum" of a player at a certain moment.

By leveraging historical quantitative model data, we have developed an advanced two-layer Long Short-Term Memory (LSTM) model. This model extensively integrates prior quantitative scores, focusing particularly on the varied impacts of different game stages on psychological stress and physiological functions. With adept transitions between short-term and long-term memory layers, the model proficiently anticipates fluctuations in athletes' momentum throughout the game. Our findings underscore the significance of thmodel's insights, leading to the following recommendations:

1. **Technical Initiative:**

   Tailor training regimes based on the model's insights, considering the varying psychological and physiological demands at different game stages. Therefore, making players pay more attention to technical initiative and creativity, and respond to the uncertainty and volatility in the game by exerting their unique playing style.

2. **Strategy Adjustment:**

   Players should flexibly adjust their game strategies based on the immediate situation of the game and the performance of the previous game, especially in the serve and return links, in order to

take advantage of the opponent's weaknesses and maximize their own advantages. This flexibility enhances overall match, helping players navigate uncertainties, seize opportunities, and contribute to resilient and successful performances.

3. **Strategic Rest and Recovery:** The coaching team should use game data analysis to identify key factors that affect game results. Also, recognize the importance of managing psychological stress and physical condition during critical game stages. Implement strategic rest and recovery plans to optimize player performance during pivotal moments, develop personalized training plans accordingly to improve players' ability to cope with game fluctuations.

We sincerely hope that our team's suggestions can be helpful to you! I wish you good results in the competition as expected!

Best,

MCM Team #2400151

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